

Hollow Soul

Current Branch Generation Reference

Runtime rules, role odds, shape weights, endpoint constraints, and reward tables. Current as of M134
Branch Pacing + Reward Room Shape Pass.

Executive read

The current beta branch is a directed, seed-driven world-loop branch. M46 supplies world-scaled target room counts, M132 supplies the active biome, M131 adds one terminal Wave Room, M133 may add one Special Encounter, M130 may add one Corrupted Chest, and M134 converts 1-2 main-route body rooms into true non-combat Reward rooms.

- Normal profile-backed world-loop branches use `m46_encounter_director_curve_v1` when the encounter catalog is present. The older M20 feature branch is the fallback path.
- The base target count includes Origin, body rooms, Secret, and Boss. Wave, Corrupted, and Special leaves are added after that base graph.
- Body-room shape selection now favors 1x1, 2x1, and 1x2. Large 2x2 and L rooms are still available but less common.
- Reward rooms are not extra leaves. They are selected from ordinary main-route body rooms and become safe cache rooms: no combat, no door lock, and no boss-key placement.
- Combat room sparse rewards remain M52-compatible. Reward rooms use a slightly richer wooden-cache roll.

Primary code sources: `BranchGenerator.cs`, `BranchSessionController.cs`, `ProceduralRewardResolver.cs`, `ChestRewardResolver.cs`.

Generation Modes

The project still contains older branch constructors, but the normal beta path is the world-loop branch created through M46 when encounter content is available. The table below explains what exists and when it matters.

Mode	Branch id	Current purpose
Legacy cross	m7_five_room_cross	Old fixed five-room cross. Used only as a last-resort fallback when macro content is unavailable.
Fixed macro branch	m14_macro_fixture_branch_v1	Macro-room fallback using the five fixture shapes directly.
Seeded macro	m15_seeded_macro_branch_v1	Seeded body placement without feature endpoints.
Feature / encounter seed branches	m17_feature_branch_v1 / m19_enemy_encounter_content_v1	Older seeded branches. May use Treasure leaf behavior when Secret is not required.
Branch features	m20_branch_features_v1	Feature branch with Secret, Boss, boss-key lock, and optional M130-M133 leaves when enabled.
Directed encounter world loop	m46_encounter_director_curve_v1	Normal beta branch path. Uses world target counts, encounter director difficulty, M132 biomes, and current optional room leaves.

Where the current run comes from

In BranchSessionController, a normal world-loop runtime is detected when the session is not spaceship or Developer Lab, macro branch content exists, branch generation settings exist, and the encounter catalog exists. That path calls the M46 directed branch generator.

Seed And Defaults

Setting	Current value	Meaning
Default seed	15001	Used when a branch seed is zero.
Default target rooms	8	Fallback base target if no world override exists.
Max placement attempts	250	Attempts per placed record before generation fails.
Loops	false	Looped graph generation is explicitly unsupported by seeded paths.
Boss leaf	true	A terminal Boss room is normally required.
Fixture shapes	1x1, 2x1, 1x2, 2x2, L3	Allowed body fixture families used for shape selection.

World	Target rooms	Difficulty offset	Base body rooms	Usually visible total
World 1	8	0	5	9 + optional rolls
World 2	10	1	7	11 + optional rolls
World 3	12	2	9	13 + optional rolls

Base body rooms = target rooms minus Origin, Secret, and Boss. The visible total assumes the guaranteed Wave Room is enabled. Corrupted and Special are independent chance leaves and may be skipped if no valid placement exists.

Generation Pipeline

Step	Pass	Rule
1	Resolve seed and content	Use the provided seed, or the default seed if zero. Select the active room pool for the current biome; fall back to the macro pool if the biome pool lacks complete shape coverage.
2	Build shape candidates	Group active rooms by footprint shape so a fixture shape can resolve to a matching biome-specific room template.
3	Place Origin	Origin is based on the 1x1 macro fixture and starts at grid coordinate (0, 0).
4	Place body rooms	Compute $\text{normalPlacementTarget} = \text{target} - 1 - \text{reserved}$. Each body room picks a parent with a free port, chooses a weighted fixture shape, then chooses a matching active-room candidate.
5	Place Secret and Boss	M20/M46 require a terminal 1x1 Secret room and a terminal Boss room. Boss may use a selected arena asset.
6	Add optional leaves	Corrupted, Wave, and Special leaves are attempted in that order. Each is terminal and excludes earlier endpoint parents.
7	Select Reward rooms	M134 selects 1 guaranteed main-route body room, plus a 50 percent seeded chance for a second, preferring smaller footprints.
8	Assign roles and IDs	Origin keeps <code>origin</code> , Boss becomes <code>boss_01</code> , all other rooms are numbered. Non-selected ordinary body rooms become Combat.
9	Connect and validate	Parent-child doors are connected, adjacent compatible ports are auto-connected, topology rules are validated, and the boss-key lock is applied for M20/M46 branches.

Shape Policy

M134 changed standard body-room selection so smaller and medium rooms dominate the default branch shape mix. The total weight is 100, so these values read directly as relative percentages before placement constraints and biome-template availability.

Footprint	Weight	Intent
1x1 single	30	Most common, fast pacing, easy reward-room candidate.
2x1 wide	25	Second most common, supports lanes and readable combat.
1x2 tall	20	Keeps vertical variety without overusing large rooms.
2x2 block	15	Medium rarity large room.
L-shape 3-cell	10	Lowest standard body-room weight; kept for layout texture.

Reward-room preference	Order
Preferred	1x1
Next	2x1
Next	1x2
Fallback	2x2
Last fallback	L-shape

Reward selection is restricted to ordinary origin-to-boss-path body rooms. Origin, Boss, Treasure, Secret, Corrupted, Wave, and Special rooms are excluded.

Room Roles

Role	Combat?	Doors	Reward behavior	Boss-key eligible?
Origin	No standard room combat	Open after start	Auto-cleared when entered in non-spaceship branches.	No
Combat	Yes	Standard combat lock behavior	Uses M52 sparse standard-room reward roll after clear.	Yes
Reward	No	Stay traversable	Immediately cleared and reward-pending. Uses M134 wooden-cache roll.	No
Boss	Yes, boss	Boss-key locked threshold	Boss reward / possible boss weapon logic.	No
Treasure	No	Stay traversable	Treasure pool or treasure-cache fallback. In M20/M46, Secret suppresses Treasure selection.	No
Secret	No	Stay traversable	Immediately cleared and reward-pending. Must be 1x1.	No
Corrupted Chest	No normal encounter	Stay traversable	Immediately reward-pending. Chest asks for confirmation and applies run max-HP loss on open.	No
Wave	Three-wave combat	Locked until final clear	Spawns Golden Chest after wave 3 clears.	No
Special Encounter	Encounter-specific	Soul Eater open; Escapist locks	Soul Eater shop offer or Escapist Golden Chest on kill.	No

Boss key rule

BranchFeaturePlan now chooses the boss-key source from Combat rooms only, ordered by farthest distance from Origin. Reward, Secret, Treasure, Corrupted, Wave, and Special rooms are excluded.

Optional Leaves

Leaf	Current policy	Topology rule	Runtime note
Secret	Guaranteed on M20/M46	Terminal 1x1	Safe reward room; boss-key branch feature uses it as the hidden feature room.
Boss	Guaranteed when boss leaf enabled	Terminal, exactly one entrance	Boss connections are marked BossKey locked in M20/M46.
Corrupted Chest	10 percent deterministic roll	At most one terminal leaf; cannot attach to Boss, Secret, or Treasure	Uses corrupted Ashen Shrine room template and Corrupted Chest pipeline.
Wave Room	Guaranteed when normal optional leaves are enabled	At most one terminal leaf; cannot attach to Boss, Secret, Treasure, or Corrupted	Three waves: 2 enemies, then 3, then 4; Golden Chest after clear.
Special Encounter	15 percent deterministic roll; kind is 50/50	At most one terminal leaf; cannot attach to Boss, Secret, Treasure, Corrupted, or Wave	Soul Eater sells a rare reward for 10 Souls; Escapist has a 20s hunt timer.

The optional leaves are enabled for normal profile-backed world-loop branches. They are not enabled for spaceship, Developer Lab, or challenge-only scaffolds.

Reward Odds

Combat rooms preserve the M52 sparse baseline. Reward rooms use a separate cache roll that is more likely to produce a Normal Chest but still can be empty.

Outcome	Combat room M52	Reward room M134
Golden Chest	2%	2%
Normal Chest	12%	30%
Loose Coins	38% - 5 to 8 coins	34% - 5 to 8 coins
HP Refill	24%	24%
Nothing	24%	10%

Chest	Contents
Normal Chest	75%: 8 to 14 coins. 25%: HP Refill.
Golden Chest	55%: 15 to 30 coins. 20%: HP Refill + 5 to 8 coins. 25%: card reward + 4 to 8 coins.
Corrupted Chest	Always 8 to 12 coins plus one curated rare build reward. Opening applies a run-scoped max HP loss.

Special source	Reward logic
Wave Room	Golden Chest after wave 3 clears.
Escapist	Golden Chest only if the Escapist is killed before the 20s timer expires.
Soul Eater	One seeded rare reward offer for 10 current-run Souls; no Banked or Unbanked copy in this prototype.
Boss	Boss reward pool, with possible boss weapon roll when the weapon reward pool is present.
Secret / Treasure	Treasure pool if present, otherwise treasure-cache fallback.

Encounter Assignment

- EncounterResolver skips Origin, Reward, Treasure, Secret, and CorruptedChest roles.
- Combat rooms use seeded or directed encounter selection, with M46 applying world difficulty offsets and capping non-boss enemy spawns at 6.
- Boss rooms use the boss encounter or boss arena selection.
- Wave rooms use RoomWaveEncounterPlan, which expands available spawn kinds into 3 runtime waves with enemy counts 2, 3, and 4.
- Special rooms use their own prototype behavior: Soul Eater is non-hostile, while Escapist is a timed hunt.

Biome And Presentation

System	Current rule
Beta world biomes	World 1: Before Teeth. World 2: The Sunken Cartouche. World 3: The Rust Choir.
Active room pool	Branch generation resolves the active biome pool and uses room templates with matching shapes. If incomplete, it falls back to the macro pool.
Corrupted rooms	Corrupted Chest room template carries corrupted_ashen_shrine and keeps that micro-biome regardless of the branch world.
Wave rooms	Wave endpoint template is wrapped with the active branch biome at runtime.
Special rooms	Soul Eater and Escapist templates are also wrapped with the active branch biome.
Chests	Normal, Golden, and Corrupted chest identities stay global for readability.

Typical Branch Shape

World	Base graph	Guaranteed extras	Chance extras	Typical read
World 1	8 rooms: Origin + 5 body + Secret + Boss	+1 Wave	10% Corrupted, 15% Special	Usually 9 rooms; expected 9.25 before placement skips.
World 2	10 rooms: Origin + 7 body + Secret + Boss	+1 Wave	10% Corrupted, 15% Special	Usually 11 rooms; expected 11.25 before placement skips.
World 3	12 rooms: Origin + 9 body + Secret + Boss	+1 Wave	10% Corrupted, 15% Special	Usually 13 rooms; expected 13.25 before placement skips.

Within the base body rooms, M134 selects 1 guaranteed Reward room plus a deterministic 50 percent chance for a second. The remaining ordinary body rooms stay Combat.

Current Design Implications

- Branch pacing now has regular safe breathers without increasing total path length: Reward rooms are converted from existing body rooms.
- Large rooms should appear less often in normal branches because their combined body-room weight is 25 out of 100.
- The branch can still spike in length and variety through optional endpoint leaves, but those leaves remain off the boss-key critical path.
- The reward economy is intentionally lean: standard combat rooms often give small or no rewards, while dedicated Reward rooms feel more cache-like.
- The minimap has role markers for Boss, Secret, Corrupted, Wave, Special, Treasure, Reward, and Origin, which keeps the added room types legible.

Source Reference

File	What it defines
Assets/_Hollow/Scripts/Hollow.Branches/BranchGenerator.cs	Branch IDs, seeded placement, optional endpoint placement, reward-room selection, topology validation, shape policy, boss-key lock.
Assets/_Hollow/Scripts/Hollow.Data/Definitions/BranchGenerationSettingsDefinition.cs	Default seed, target count, allowed fixture IDs, placement attempts, boss/treasure flags.
Assets/_Hollow/Scripts/Hollow.Data/Definitions/EncounterDirectorProfileDefinition.cs	M46 world target counts, difficulty offsets, and non-boss spawn cap.
Assets/_Hollow/Scripts/Hollow.Branches/BranchRoomRole.cs	Current role enum: Origin, Combat, Reward, Boss, Treasure, Secret, CorruptedChest, Wave, SpecialEncounter.
Assets/_Hollow/Scripts/Hollow.Branches/BranchFeaturePlan.cs	Boss-key source selection from Combat rooms only.
Assets/_Hollow/Scripts/Hollow.Branches/EncounterResolver.cs	Encounter assignment skip roles and directed encounter planning.
Assets/_Hollow/Scripts/Hollow.Branches/BranchSessionController.cs	Runtime entry behavior, room clear/reward-pending state, world-loop feature enablement, M46 graph creation.
Assets/_Hollow/Scripts/Hollow.Branches/ProceduralRewardResolver.cs	M52 combat reward odds, M134 reward-room cache odds, Wave/Special/Corrupted reward grants.
Assets/_Hollow/Scripts/Hollow.Rewards/ChestRewardResolver.cs	Normal, Golden, and Corrupted chest contents and corrupted roll percent.
Assets/_Hollow/Scripts/Hollow.Branches/SpecialEncounterResolver.cs	Special Encounter roll percent, 50/50 kind selection, Soul Eater price, Escapist timer.
Assets/_Hollow/Scripts/Hollow.Combat/RoomWaveEncounterPlan.cs	Wave split policy: 2, 3, and 4 enemies.
Assets/_Hollow/Scripts/Hollow.Branches/BranchSessionContent.cs	Biome room-pool resolution and biome wrapping for Wave/Special templates.

Useful testing lens

For a branch seed scan, inspect role counts, footprint classes, boss-key room role, and optional leaf terminal status. The key current expectation is: 1-2 Reward rooms on the main route, one Wave leaf when enabled, no boss-key outside Combat, and fewer 2x2/L body rooms than small or medium footprints.